

Loops (for loop)

for loop

Structure of **for loop**:

```
1. for ( INITIALIZATION ; CONDITION ; STEPS ) {  
2.     BODY OF THE LOOP  
3. }
```

Key points of the structure:

- We **initialize** variable, write the **condition** and **increase/decrease** the initialized variable within a parenthesis right after the **for** keyword.
- The aforementioned 3 components are separated using ; (**semicolon**) .
- We can initialize or increase/decrease multiple variables at once by separating them using , (**Comma**) .

for loop (contd.)

```
1. public class ForLoopScenario1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 500 ; i++ ) {
4.             System.out.println( "CSE is fun" );
5.         }
6.     }
7. }
```

Scenario 1: Write a java program using for loop to print "CSE is fun" 500 times.

Welcome to DrJava. Working directory is /home/Java110

> run ForLoopScenario1

CSE IS FUN

CSE IS FUN

.....

for loop vs while loop (contd.)

```
1. public class WhileLoopScenario1{
2.     public static void main(String[] args) {
3.         int i = 0;
4.         while ( i < 500 ) {
5.             System.out.println("CSE IS
6.             FUN");
7.             i++;
8.         }
9.     }
```

Initialization points to line 3.

Condition points to line 4.

Body points to line 5.

Steps points to line 6.

```
1. public class ForLoopScenario1{
2.     public static void main(String[] args) {
3.         for ( int i = 0 ; i < 500 ; i++ ) {
4.             System.out.println( "CSE is fun" );
5.         }
6.     }
7. }
```

Welcome to DrJava. Working directory is /home/Java110

> run `ForLoopScenario1`

CSE IS FUN

CSE IS FUN

.....

Welcome to DrJava. Working directory is /home/Java110

> run `ForLoopScenario1`

CSE IS FUN

CSE IS FUN

.....

for loop (contd)

It is also possible to go backward using for loop.
Let's say we want to print from 50 to 1.

```
1. public class ForLoopUltra{
2.     public static void main(String[] args) {
3.         for ( int i = 50; i >= 1; i-- ) {
4.             System.out.print(i + " ");
5.         }
6.     }
7. }
```

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> run ForLoopUltra

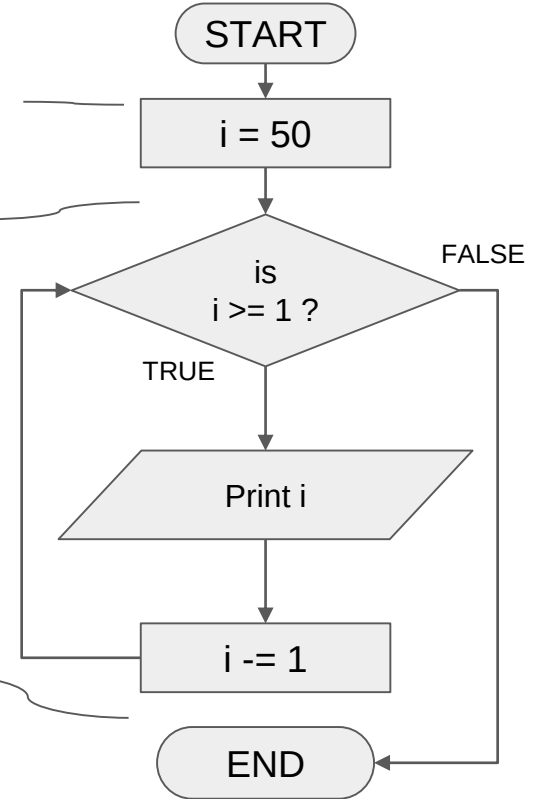
```
50 49 48 47 46 45 44 43 42 41 40 39 38 37 36 35 34 33 32 31 30 29
28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5
4 3 2 1 >
```

Initialization

Condition

Body

Steps



for loop (Contd)

Scenario 2 flowchart vs Java code using for loop.

```
1. public class ForLoopScenario2{
2.     public static void main(String[] args) {
3.         for (int i = 0; i <= 120; i++) {
4.             if (i % 7 == 0) {
5.                 System.out.print( i + " ");
6.             }
7.         }
8.     }
9. }
```

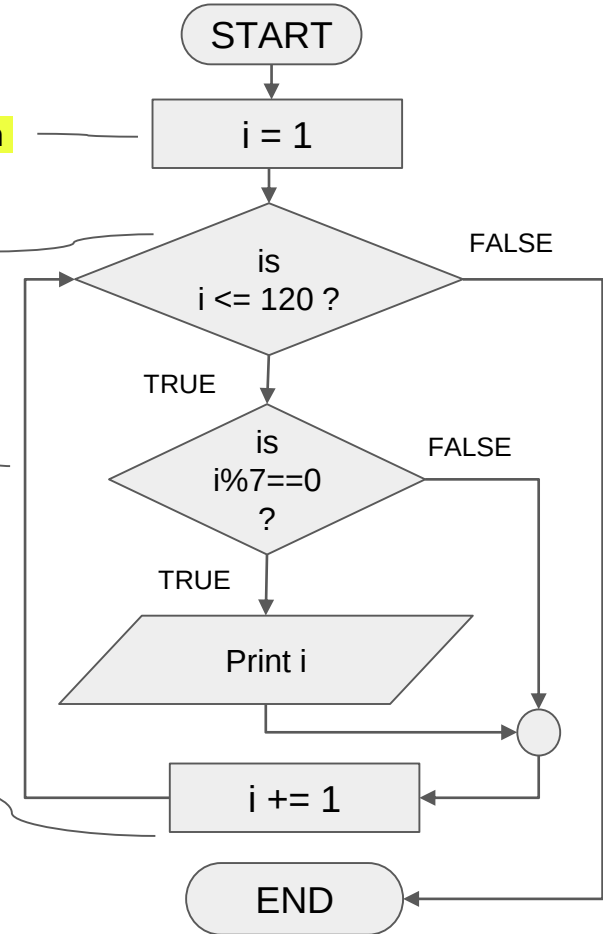
Welcome to DrJava. Working directory is /home/Java110
> run ForLoopScenario2
7 14 21 28 35 42 49 56 63 70 77 84 91 98 105 112 119 >

Initialization

Condition

Body

Steps



THANK YOU!